CHAPTER

TWO

Formulation and Numerical Method

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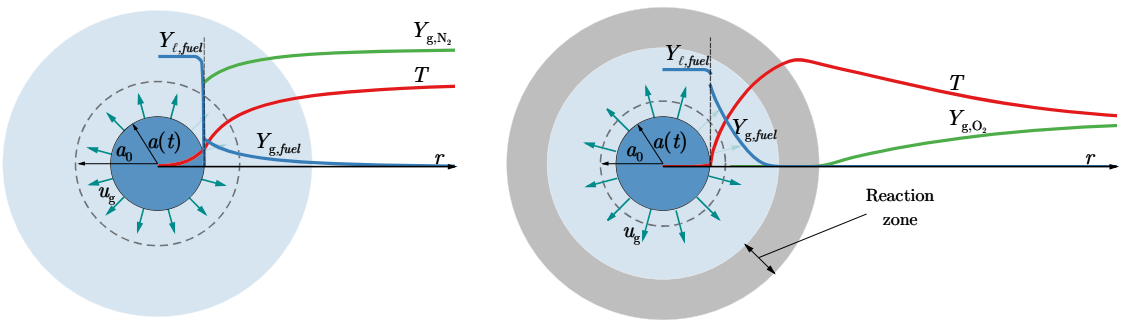
2.1 Introduction

Droplet ignition and combustion is a very relevant problem in combustion applications, what explains the existence of a multitude of theoretical [1–4] and experimental [5–8] studies explaining different aspects of the problem. In view of the increasing use of biofuels, either as pure fuels or mixed with other hydrocarbons, the demand of accurate modelling of the vaporization and combustion of

such fuels is expected to grow in the close future. During the literature survey carried out in this thesis, we only found a few numerical codes capable of tackling the problem. It is worth mentioning the work by Marchese et al [9, 10], capable of computing the quasi-steady stage of the combustion of multicomponent droplets using detailed chemical schemes, or the works by Cuoci et al. [11] and Stauch et al. [12], suited to compute the full transient evaporation and combustion of monocomponent droplets.

The aim of this chapter is to develop a numerical code, based on a conservative discretization of the equations derived from first principles, capable of dealing with the unsteady evaporation and combustion of multicomponent droplets.

2.2 Problem formulation



(a) Pure vaporization case **(b)** Complete combustion case

Figure 2.1: Sketch of the spherically-symmetrical droplet vaporization and combustion problems, subfigures (a) and (b) respectively.

We will consider here the case of a single droplet with initial diameter of $d_0 = 2a_0$ in an infinite stagnant ambient without gravity or forced flow as shown in the sketch in Fig. 2.1. In these conditions, the flow has spherical symmetry provided that the initial conditions satisfy this property.

2.2.1 Conservation equations

The transient, spherically symmetric, problem of droplet evaporation and combustion in microgravity conditions is mathematically described by the conservation

of mass, species and energy,

$$\frac{\partial \rho_\beta}{\partial t} + \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 \rho_\beta u_\beta) = 0, \tag{2.1}$$

$$\frac{\partial(\rho_\beta Y_{\beta,i})}{\partial t} + \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 \rho_\beta Y_{\beta,i} u_\beta) = -\frac{1}{r^2}\frac{\partial}{\partial r}(r^2 J_{\beta,i}) + \omega_{\beta,i}, \quad i = 2, ..., N_\beta, \tag{2.2}$$

$$\frac{\partial(\rho_\beta h_\beta)}{\partial t} + \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 \rho_\beta h_\beta u_\beta) = -\frac{1}{r^2}\frac{\partial}{\partial r}(r^2 q_\beta) + Q_\beta, \tag{2.3}$$

with ρ_β, u_β and h_β the density, the velocity and thermal enthalpy of the mixture, respectively. The sub-index i represents the ith species in both the liquid phase $\beta = \ell$ inside the droplet and the gas phase $\beta = \mathrm{g}$. $Y_{\beta,i}$ is the mass fraction and $Q_\beta = -\sum_{i=1}^{N_\beta} h^{\mathrm{ref}}_{\beta,i}\omega_{\beta,i}$ the heat released, with $h^{\mathrm{ref}}_{\beta,i}$ the enthalpy at the reference temperature $T^{\mathrm{ref}} = 298.15$ K. Heat production in the liquid phase is neglected. The mass production or consumption rates $\omega_{\beta,i}$ [1] are calculated using an appropriate chemical kinetic mechanism formed by N_β species and N_Ω reactions.

The evaluation of the species conservation equations and reaction rates constitute a great part of the total computational cost of the problem, that increases as N_β^2 for explicit solvers or as N_β^3 in implicit solvers because of the calculation of the Jacobian matrix [13]. Typically, the solution of complex systems using detailed combustion schemes is extremely expensive from a computational point of view, reason why it is common the use of reduced combustion mechanisms that are usually developed to describe specific aspects of combustion. Unfortunately, the problem at hand is very complex and requires of a reduced mechanism capable of describing, accurately and efficiently, autoignition, steady and unsteady premixed and diffusion flames and extinction, for a range pressure, temperature and equivalence ratio relevant for practical applications. Such a multipurpose reduced mechanism was not available in the literature and is developed in Chapter 4 to show that the computational cost can be reduced between 11 and 38 times, depending on the numerical algorithm used.

The species mass flux term for the gas phase in equation (2.2) is calculated with the mixture averaged model [14] with conservative flux correction [15]

$$J_{\mathrm{g},i} = -\rho_{\mathrm{g}} Y_{\mathrm{g},i}\left(V^0_{d,i} + V^c_d\right), \tag{2.4}$$

where $V^0_{d,i} = -(D_{\mathrm{g},i}/X_{\mathrm{g},i})(\partial X_{\mathrm{g},i}/\partial r)$, $V^c_d = -\sum_{i=1}^{N_{\mathrm{g}}} Y_{\mathrm{g},i} V^0_{d,i}$, $X_{\beta,i}$ is the mole fraction and $D_{\mathrm{g},i} = (1 - Y_{\mathrm{g},i})/(\sum_{j\neq i}^{N_{\mathrm{g}}} X_{\mathrm{g},j}/D_{\mathrm{g},ji})$ is the mixture diffusion coefficient, with $D_{\mathrm{g},ij}$ the binary diffusion coefficient, for the pair of species i and j, obtained from the kinetic theory [16]. For the liquid phase, the mass flux term is

[1] For an elementary reaction as $\nu_{\mathrm{A}}\mathrm{A} \rightarrow \nu_{\mathrm{B}}\mathrm{B}$, the mass consumption rate of specie A takes the form $\omega_{\beta,\mathrm{A}} = -\nu_{\mathrm{A}} W_{\mathrm{A}} C_{\mathrm{A}}^{\nu_{\mathrm{A}}}\kappa$ where ν_A is the stoichiometric coefficient of specie A in the reaction, C_A is the mole concentration and κ is the reaction rate constant for the reaction.

calculated from Fick's law as

$$J_{\ell,i} = -\rho_\ell D_{\ell,i} \frac{\partial Y_{\ell,i}}{\partial r} \tag{2.5}$$

and the mixture diffusion coefficient is determined by the Wilke-Chang equation [17]

$$D_{\ell,i} = 1.173 \times 10^{-16} \frac{\sqrt{\sum_{j\neq i}^{N_\ell} X_{\ell,j}\varphi_j W_j T}}{\mu_\ell \mathrm{V}_{\ell,i}^{0.6}} \tag{2.6}$$

where W_i is the species mole mass of the liquid phase, μ_ℓ the mixture viscosity, $\mathrm{V}_{\ell,i}$ the molar volume. The association factor takes the values $\varphi_i = 2.6$ for water and $\varphi_i = 1.5$ for ethanol. In equation (2.3), the thermal heat flux term q_β is obtained from the generalized Fourier's law,

$$q_\beta = -k_\beta \frac{\partial T}{\partial r} + \sum_{i=1}^{N_\beta} J_{\beta,i} h_{\beta,i}, \tag{2.7}$$

where k_β is the thermal conductivity of the β phase and N_β denotes the number of species on each phase. Soluble species exist in both phases ($i = 1, ..., N_\ell$) but non-soluble species only exists in the gas phase ($i = N_\ell + 1, ..., N_\mathrm{g}$). Only major liquid solubilities such as water-ethanol or alkane-alkane have been considered avoiding minor ones as those of nitrogen or carbon dioxide in the liquid phase. The most abundant species in each phase, ($i = I$), are obtained as $Y_{\beta,I} = 1 - \sum_{i\neq I}^{N_\beta} Y_{\beta,i}$. Equations (2.1)-(2.7) are supplemented with the Equations of State (EoS) of the gas written in the quasi-isobaric approximation $p_\infty/\rho_\mathrm{g} = TR_\mathrm{g}$. The density of the mixture in the liquid phase is computed as in [18] using the expression $\rho_\ell = \left(\sum_{i=1}^{N_\ell} X_{\ell,i}\rho_{\ell,i}^{1/2}\right)^2$, in which the individual densities of liquid species are obtained by fitting published experimental data using the expression introduced by [19],

$$\log \rho_{\ell,i} = A_{\rho,i} \log(T) + \frac{B_{\rho,i}}{T} + \frac{C_{\rho,i}}{T^2} + D_{\rho,i} + E_{\rho,i}T + F_{\rho,i}T^2 \tag{2.8}$$

2.2.2 Constitutive relations

Both the liquid and gas phases are considered ideal mixtures with heat capacity and enthalpy calculated as $c_{p_\beta} = \sum^{N_\beta} Y_{\beta,i} c_{p_{\beta,i}}$ and $h_\beta = \sum^{N_\beta} Y_{\beta,i} h_{\beta,i}$, in terms of the heat capacity $c_{p_{\beta,i}}$ and thermal enthalpy $h_{\beta,i}$ of the species i in the phase β.

The thermodynamic properties $c_{p_{\beta,i}}$ and $h_{\beta,i}$ of pure species are obtained using the NASA polynomials, where the coefficients are obtained, whenever possible, from the San Diego mechanism database [20]. Those species not available in the

San Diego database were taken from Burcat's database [21] or, if not available in the literature, by fitting experimental data.
The gas phase molecular transport coefficients $D_{\mathrm{g},ij}$ are obtained using the expression derived directly from the kinetic theory [16] using the transport database of the San Diego mechanism [20], while k_{g} is obtained using standard mixture average formula [22],

$$k_{\mathrm{g}} = \frac{1}{2}\left(\sum_{i=1}^{N_{\mathrm{g}}} X_{\mathrm{g},i} k_{\mathrm{g},i} + \frac{1}{\sum_{i=1}^{N_{\mathrm{g}}} X_{\mathrm{g},i}/k_{\mathrm{g},i}}\right). \tag{2.9}$$

For the liquid phase, the mixture thermal conductivity k_ℓ was obtained from a generalization of Filippov's equation [23],

$$k_\ell = \sum_{i=1}^{N_\ell} Y_{\ell,i}\left(k_{\ell,i} - \sum_{j=i+1}^{N_\ell} K_{i,j} Y_{\ell,j}\left|k_{\ell,i} - k_{\ell,j}\right|\right) \tag{2.10}$$

where Filippov's constant is $K_{i,j} = 0.72$. The conductivity of the pure species is computed using the correlation [19],

$$\log k_{\ell,i} = A_{k,i}\log(T) + \frac{B_{k,i}}{T} + \frac{C_{k,i}}{T^2} + D_{k,i} + E_{k,i}T + F_{k,i}T^2, \tag{2.11}$$

in which the coefficients are obtained by fitting experimental data available in the literature. The viscosity of the liquid mixture is evaluated using the Grunberg and Nissan equation [24]

$$\mu_\ell = \exp\left(\sum_{i=1}^{N_\ell} X_{\ell,i}\ln\mu_{\ell,i}\right) \tag{2.12}$$

where the viscosity of the pure species is obtained using an expression analogous to Eq. (2.11). The fitting coefficients obtained for liquid specie properties and the corresponding experimental data source, are collected in appendix 2.B.

2.2.3 Boundary conditions

Boundary conditions are required at the center of the droplet and in the far field,

$$r = 0 : \frac{\partial T_\ell}{\partial r} = \frac{\partial Y_{\ell,i}}{\partial r} = u_\ell = 0, \tag{2.13}$$

$$r \to \infty : T_{\mathrm{g}} - T_\infty = Y_{\mathrm{g},i} - Y_{\infty,i} = 0, \tag{2.14}$$

while the boundary conditions at the liquid-gas interface are obtained by imposing the conservation of species mass and energy in a control volume extending from $r = a(t) - \delta$ to $r = a(t) + \delta$ in the limit $\delta \to 0$, yielding

$$-\dot{m}''(Y_{\mathrm{g},i} - Y_{\ell,i})_{r=a} = -(J_{\mathrm{g},i} - J_{\ell,i})_{r=a}, \qquad i = 2, ..., N_\ell, \tag{2.15}$$

$$-\dot{m}''(Y_{\mathrm{g},i})_{r=a} = -(J_{\mathrm{g},i})_{r=a}, \qquad i = N_\ell + 2, ..., N_\mathrm{g}, \tag{2.16}$$

$$-\dot{m}'' \sum_{i=1}^{N_l} \Big(Y_{\ell,i}\mathrm{L}_i(T)\Big)_{r=a} = \left(k_\mathrm{g}\frac{\partial T}{\partial r} - k_\ell \frac{\partial T}{\partial r}\right)_{r=a} + \alpha_\mathrm{eff}\sigma\left(T_\infty^4 - T_s^4\right) - \sum_{i=1}^{N_l}\Big(J_{\ell,i}\mathrm{L}_i(T)\Big)_{r=a}, \tag{2.17}$$

where $a(t)$ is the instantaneous time-dependent radius of the droplet at a generic time t,

$$\dot{m}'' = -\rho_\ell(u_\ell - \dot{a})_{r=a} = -\rho_\mathrm{g}(u_\mathrm{g} - \dot{a})_{r=a} \tag{2.18}$$

is the mass vaporization rate per unit of surface area, T_s is droplet surface temperature respectively and $\dot{a} = da/dt$. α_eff is the effective radiation absorption coefficient and σ is the Stefan-Boltzmann constant. The vaporization heat of each species is calculated as $\mathrm{L}_i(T) = h_{\mathrm{g},i}(T) - h_{\ell,i}(T) + \mathrm{L}_i^\mathrm{ref}$, with $\mathrm{L}_i^\mathrm{ref}$ representing the vaporization heat at the reference temperature.
Additionally, imposing the conservation of the chemical potential at the interface we obtain the Clausius equation [25],

$$(Y_{\mathrm{g},i})_{r=a} = \left(Y_{\ell,i}\frac{W_\ell}{W_\mathrm{g}}\right)_{r=a} \frac{p_\mathrm{atm}}{p_\infty}\gamma_i \exp\left(\int_{T_{\mathrm{b},i}}^{T_s} \frac{\mathrm{L}_i(T)}{R_{\mathrm{g},i}T^2}dT\right), \quad i = 1, ..., N_\ell, \tag{2.19}$$

where $T_{\mathrm{b},i}$ is the boiling temperature at atmospheric pressure ($p_\mathrm{atm} = 101325\mathrm{Pa}$), $R_{\mathrm{g},i}$ is the specific gas constant and γ_i the activity coefficient for the i species, obtained using the UNIFAC method [26], see appendix 2.A for details.

2.2.4 Initial conditions

Suitable initial conditions are required in order to pose the time-dependent problem. We will consider here the simplest case of a droplet with uniform composition and temperature that is placed at $t = 0$ in an infinitely large homogeneous gaseous ambient. Both temperature and fuel mass fraction profiles would then be discontinuous at the interface. In order to reduce the rigidity of such initial condition, the problem is integrated in time from $t = t_0$, using as initial condition the an-

alytical solution obtained by Sazhin et al. [27] for an isolated non-evaporating droplet, to generate a smooth initial temperature profile

$$t = t_0 : \; T_{\mathrm{g}} = T_\infty + \frac{a_0}{r}(T_{d_0} - T_\infty)\mathrm{erfc}\left(\frac{r - a_0}{2\sqrt{D_{T,\mathrm{g},\infty}\; t_0}}\right), \tag{2.20}$$

valid for $t_0 \ll a_o^2/D_{T,\mathrm{g}}$, with $D_{T,\mathrm{g},\infty} = k_{\mathrm{g},\infty}/c_{p_{\mathrm{g},\infty}}\rho_{\mathrm{g},\infty}$ being the thermal diffusivity at infinity. The numerical parameter t_0 is as small as 10^{-5}s to minimize its influence on the posterior evolution of the problem, and sufficiently large to reduce the numerical stiffness of the problem. To avoid numerical problems, we imposed a smooth initial profile for the species mass fraction inspired by eq. (2.20) and defined as

$$t = t_0 : \; Y_{\mathrm{g},i} = Y_{\mathrm{g},i,\infty} + \frac{a_0}{r}(Y_{\mathrm{g},i,0} - Y_{\mathrm{g},i,\infty})\mathrm{erfc}\left(\frac{r - a_0}{2\sqrt{D_{T,\mathrm{g},\infty}\; t_0}}\right) \tag{2.21}$$

where the initial vaporized species profile at the droplet surface, $Y_{\mathrm{g},i,0}$, is a small quantity, of the order of 10^{-8}, that reduces the rigidity of the step-function initial condition but does not affect the solution.

To complete the set of initial condition, we assume that at $t = t_0$ the droplet, of uniform temperature and composition, is suddenly placed unperturbed in the gaseous ambient,

$$t = t_0 : \; Y_{\ell,i} - Y_{\ell,i,0} = T_\ell - T_{d_0} = 0; \quad i = 1, ..., N_\ell \tag{2.22}$$

2.2.5 The Rapid-Mixing limit

The formulation given above in (2.1)-(2.3) constitutes the full 1D model and takes into account the existence of radial gradients in the concentration of ethanol and water in the liquid phase. As a simpler alternative, valid when the liquid phase homogenization $a_0^2/D_{T,l}$ time is much smaller than the droplet life-time $\tilde{t}_{\mathrm{evap}} \sim \left(4\pi a_0^3\rho_\ell/3\right)/\left(4\pi a_0^2\dot{m}''\right)$, the temperature and composition in the droplet can be taken as homogeneous. This limit can be used in those cases in which internal recirculation within the droplet exists as a consequence of forced, natural or Marangoni convection, induced either by temperature gradients or by the fibers used to maintain the droplet at a fixed position. The Rapid-Mixing (RM) limit, originally described by Law [28], was considered during the calculations shown in the next chapters to learn about the limits of applicability of such simplification. When the RM limit is used, the problem in the liquid phase is simplified Eqs. (2.2) and (2.3) can be integrated for the whole droplet, that becomes a single control

volume reducing the problem to the integration of the homogeneous equations

$$\frac{\mathrm{d}}{\mathrm{d}t}(\rho_\ell Y_{\ell,i} V_d) = \Big(\dot{m}''(Y_{\mathrm{g},i})_{r=a} - (J_{\mathrm{g},i})_{r=a}\Big) A_d, \quad i = 2, ..., N_\ell \tag{2.23}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(\rho_\ell h_\ell V_d) = \Big(\dot{m}''(h_{\mathrm{g}})_{r=a} - (q_{\mathrm{g}})_{r=a} + \alpha_{\mathrm{eff}}\sigma(T_\infty^4 - T_s^4)\Big) A_d, \tag{2.24}$$

where V_d and A_d are, respectively, the droplet volume and surface area at time t.

2.3 Numerical method

The problem formulated in section 1.2 involves the solution of the conservation equations in two phases, coupled through the boundary conditions at the droplet surface. The temporal evolution of the droplet radius $a(t)$, where boundary conditions are applied, has to be determined in the course of the solution. In this section we will describe the discretization and the different algorithms used in the tracking of the free surface and in the numerical solution of the discretized equations. Given the complexity of the flow, computational efficiency is required if solution is to be obtained in reasonable times.

2.3.1 Numerical discretization of the gas and liquid phases

A cell-centered finite volume method has been employed to discretize the governing equations. For an arbitrary control volume, equations (2.1)-(2.3) take the general form

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{V_c(t)} \phi_\beta \mathrm{d}V + \int_{S_c(t)} \phi_\beta(\bar{u}_\beta - \bar{v}_c)\cdot\bar{n}\mathrm{d}S = \int_{S_c(t)} \bar{\mathscr{F}}(\phi_\beta)\cdot\bar{n}\mathrm{d}S + \int_{V_c(t)} \mathscr{P}(\phi_\beta)\mathrm{d}V \tag{2.25}$$

where ϕ_β is any conservative variable, $\phi_\beta = \{\rho_\beta,\ \rho_\beta Y_{\beta,i},\ \rho_\beta h_\beta\}$, where $i = \{2, ..., N_\beta\}$, $V_c(t)$ is the control volume delimited by the control surface $S_c(t)$, $\bar{u}_\beta$ and $\bar{v}_c(t)$ are the local velocities of the fluid and control surface, respectively, and $\bar{\mathscr{F}}(\phi_\beta) = \{0,\ \bar{J}_{\beta,i},\ \bar{q}_\beta\}$ and $\mathscr{P}(\phi_\beta) = \{0,\ \omega_{\beta,i},\ Q_\beta\}$ are the diffusive flux and volumetric production rate, respectively, for the conservative variable ϕ_β. In our case, the domain is discretized in a set of concentric, spherically-symmetrical control volumes in which each volume V_n is bounded by two concentric spherical surfaces S_n- and S_n+, as shown in figure 2.2. Notice that the control volumes are concentric and do not overlap, $S_{n-} = S_{(n-1)+}$. The innermost, gas-phase control surface coincides with the outermost, liquid-phase control surface and they both lie at the droplet interface. Because the radius $a(t)$ of the droplet is moving with a velocity $\mathrm{d}a/\mathrm{d}t$, this control surface moves as well. To avoid the control volumes from collapsing or expanding excessively, all control volumes and their

corresponding control surfaces are continuously adapted to the droplet size, with the recession velocity $\bar{v}_c$ in (2.25) obtained using values of $a(t)$ calculated in next time steps.

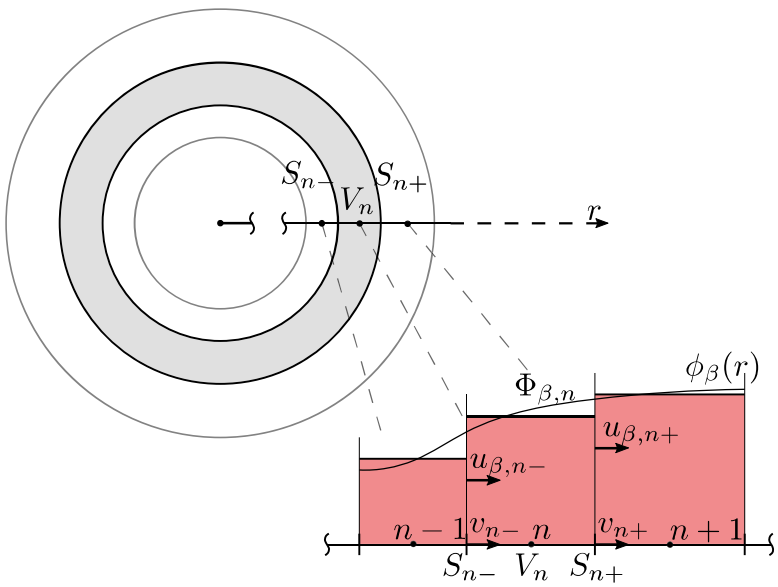


Figure 2.2: Sketch of spherical symmetric finite volume $V_n(t)$ delimited by two spherical surfaces $S_{n-}(t)$ and $S_{n+}(t)$ of radii $r_{n-}(t)$ and $r_{n+}(t)$, respectively; $n\pm$ subscripts are employed to designed the surfaces adjacent to the node n, where $r_{n-1+}(t) = r_{n-}(t)$. The cell-centered arrangement with the staggered grid for the velocity field is shown below; the cell-average values $\Phi_{\beta,n} = \phi_{\beta,n} + \mathcal{O}(h^2)$ are stored at the cell centers, and the velocities of the fluid and control surfaces, $u_{\beta,n\pm}$ and $v_{n\pm}$ respectively, are stored at the control volumes interfaces.

Integration of equation (2.25) over a spherically symmetric finite volume V_n, as sketched in figure 2.2, yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(\Phi_{\beta,n}V_n\Big) + \Big[\phi_\beta(u_\beta - v_c)A\Big]_{n-}^{n+} = -\Big[\mathscr{F}(\phi_\beta)A\Big]_{n-}^{n+} + \mathscr{P}(\Phi_{\beta,n})V_n, \tag{2.26}$$

where $n = \{1, ..., \mathrm{n}_\beta\}$, being n_β the total number of nodes in the β phase, and

$$\Phi_{\beta,n} = \frac{\int_{V_n(t)} \phi_\beta \mathrm{d}V}{V_n(t)} \tag{2.27}$$

is the volume-average of ϕ_β, over the volume V_n. Notice that, consistenly with the order of the discretization, $\Phi_{\beta,n} = \phi_{\beta,n} + \mathcal{O}(h^2)$, being h the local distance between grid points.

The convective fluxes of the variable ϕ_β through the surfaces $S_{n\pm}(t)$, are evaluated from the net velocity of the fluid relative to the surface $u_{\beta,n\pm} - v_{c,n\pm}$, the area of the surface $A_{n\pm}$ and the local value of $\phi_{\beta,n\pm}$ at the surfaces, obtained

from interpolation from the neighbour volumes. A monotone first order upwind interpolation has been used to obtain these values, according to

$$\phi_{\beta,n+} = \begin{cases} \Phi_{\beta,n} + \mathcal{O}(h), & u_{\beta,n+} > 0 \\ \Phi_{\beta,n+1} + \mathcal{O}(h), & u_{\beta,n+} < 0 \end{cases}. \tag{2.28}$$

Notice that this upwind interpolation is also the simplest TVD (Total variation Diminishing) method. Higher order TVD methods, based on the introduction of flux-limiters [29] to avoid numerical oscillations, were tested, but their use complicates the algorithms and were found not to be necessary in this work.

The evaluation of the diffusive fluxes require the calculation of gradients at the boundaries of each control volume n. This can be easily achieved with a second order discretization, from the values of the variables at the centers of the neighbour volumes n and $n+1$

$$\left.\frac{\partial \phi}{\partial r}\right|_{n+} = \frac{\phi_{n+1} - \phi_n}{r_{n+1} - r_n}, \tag{2.29}$$

that recovers the well known second order centered approximation for $\mathrm{d}\phi/\mathrm{d}r$, as $n+$ is located midway between nodes n and $n+1$.

2.3.2 The numerical treatment of the continuity equation

The mass conservation equation (2.1) plays a fundamental role in ensuring both accuracy and stability of the numerical method. In the 1-dimensional problem we are considering here, the velocity is obtained from the continuity equation and the momentum equation doesn't need to be integrated unless the pressure field is required. If, as explained below, a backward Euler implicit method is used to discretize the unsteady terms, the velocity field can be obtained from the discretization of equation (2.1),

$$\left(\rho_\beta u_\beta A\right)_{n+}^{t+\Delta t} = \left(\rho_\beta u_\beta A\right)_{n-}^{t+\Delta t} + \left[\left(\rho_\beta v_c A\right)^{t+\Delta t}\right]_{n-}^{n+} - \frac{1}{\Delta t}\left[\rho_{\beta,n} V_n\right]_t^{t+\Delta t}, \tag{2.30}$$

that yields $u_{\beta,n+}^{t+\Delta t}$ as a surrogate variable from the conservative variables $\Phi_{\beta,n}^{t+\Delta t}$ through $\rho_\beta^{t+\Delta t}$. The continuity equation transmits the information instantaneously to the whole domain and becomes, then, a constraint that needs to be satisfied in order to guarantee mass conservation and to ensure a good convergence of the numerical method. The solution calculated following this procedure is analogous to solving the momentum equation to obtain the velocity field to, later, transform the continuity equation into a Poisson-like equation for the pressure. The expression $\left(\rho_{\beta,n}^{t+\Delta t} V_n^{t+\Delta t} - \rho_{\beta,n}^{t} V_n^{t}\right)/\Delta t$ accounts for the volume change in the control volumes under consideration and it can be calculated once the rest

of variables are already known $\rho_\beta h_\beta$, $\rho_\beta Y_{\beta,i}$.

2.3.3 The droplet surface tracking method and the meshing strategy

The discretization errors in the different terms of the equations are $\mathcal{O}(h^2)$, except for the convective flux, which is $\mathcal{O}(h)$ and, therefore, the overall accuracy is of order $\mathcal{O}(h)$ in the limit $h \to 0$ in a uniform grid. The problem has multiple scales, in both space and time, and the computational cost associated with the use of a uniform grid is too great. Instead, we used a non-uniform grid where the computational nodes are concentrated in regions in which the steep gradients in the variables make necessary additional accuracy.

The physical domain extends from $r = 0$ to $r = \infty$ an is divided in three regions. Near the droplet surface and near the flame, the grid is uniformly spaced to properly describe the sharp changes expected in the numerical values of the variables, especially the temperature and the species mass fractions. The spacing between the gird nodes is gradually increased in other regions where the gradients of the variables are not that steep.

The first region is formed by liquid phase and is discretized using a uniform grid with $\mathrm{n}_\ell = 20$ points. The gas-phase adjacent to the droplet constitutes the second region. It starts at the droplet surface and is sufficiently large to encompass all the relevant phenomena taking place in the gas phase -heat and mass exchange with the droplet, autoignition and quasi-steady combustion. This region extends from $r = a(t)$ to $r = r_\mathrm{I} = 5a_0$ and is discretized using a uniform grid spacing to minimize the loss of accuracy associated to the numerical discretization of non-uniform grids. The total number of grid nodes allocated in this region is about $\mathrm{n}_{\mathrm{g,I}} = 60$.

The third zone extends from $r = r_\mathrm{I}$ to the outer boundary $r = r_\mathrm{end}$ of a computational domain that is made sufficiently large to ensure border effects, with r_end ranging from $r_\mathrm{end} = 100a_0$ and $300a_0$, depending on the case considered. In this region the gradients of all magnitudes of interest are small and larger distance between the nodes is allowed. The mesh is built with a constant expansion ratio $\alpha > 1$, such that $\Delta r_{n+1} = \alpha \Delta r_n$, starting from the uniform grid space of region 2. The parameter α is calculated using the number of nodes specified for this region $\mathrm{n_g} - \mathrm{n_{g,I}}$ and imposing that the last node is located exactly at $r = r_\mathrm{end}$. Typically $\mathrm{n_g} = 120$ is sufficient to obtain an accurate solution to the problem.

To track the position of the gas-liquid interface we implemented a moving mesh method [30] that makes use of a two-step, predictor-corrector strategy. Once the new value of the variables are known, the new position of the interface is computed by integrating $\dot{a} = \dot{m}''/\rho_\ell + (u_\ell)_{r=a}$. After that, we define a new grid evolving

the grid points to a new location using the velocity of the nodes $v_n = \dot{a}r_n/a$. After that, the variables are again calculated in the new grid and the procedure continues until the normalized difference between the interface velocity calculated in two consecutive iterations falls below 10^{-5}, as indicated in Fig. 2.3.

2.3.4 The algorithms for non-linear equations

The non-linear algebraic equations system

The system of equations given above in Eqs. (2.1)-(2.3), the equation of state (EoS), the definition of the dominant species, the boundary conditions at the center of the droplet and at the end of the domain (2.13)-(2.14) and, finally, the interface equations (2.15)-(2.19) which couple both phases and require the corresponding the interface variables, $\phi_s = \{\rho_{s,\ell},\ Y_{s,\ell,i},\ T_s,\ \rho_{s,\mathrm{g}},\ Y_{s,\mathrm{g},i},\ \dot{m}''\}$, $i = \{2, ..., N_\beta\}$, determine the numerical problem to be solved.

Typically, in chemically reacting problems, the reaction terms are characterized by their strong non-linearity and numerical stiffness. Explicit schemes, such as Runge Kutta methods, require typically very small time steps because of numerical stability constraints. Linear stability restrictions to the time step can be avoided using an implicit method, where a variety of methods, of different accuracy, exist in the literature. We have used Euler's backward differentiation formula because of its simplicity and robustness in spite of its relatively low order accuracy (it introduces discretization errors of order $\mathcal{O}(\Delta t)$). The short characteristic time of the physical phenomena to be described (of the order of $\Delta t \sim 10^{-5}$s at the sudden autoignition event) and the non-linear stability constraints posed by the problem, cancel the advantage of using higher order numerical methods that would allow larger time steps [31].

If the unsteady terms in the conservation equations are discretized using Euler's method, the numerical problem to be solved for each time step reduces to finding the unknowns at $t + \Delta t$ that cancel the residuals function formed by substracting the left and right hand side of Eqs. (2.1)-(2.3), yielding

$$\tilde{f}^{t}_{0\beta,n} = \frac{-1}{\Delta t}\Big[\Phi_{\beta,n}V_n\Big]_{t}^{t+\Delta t} - \Big[\Big(\phi_\beta(u_\beta - v_n)A\Big)\Big]_{n-}^{n+} - \Big[\mathscr{F}(\phi_\beta)A\Big]_{n-}^{n+} + \mathscr{P}(\Phi_{\beta,n})V_n. \quad (2.31)$$

It is convenient, before the description of the method, to introduce the notation. All variables at time t are grouped in a single vector of unknowns Ω^t and all the

residuals $\tilde{f}^t_{0\beta,n}$ are grouped in a residuals vector $\tilde{f}^t_0$, defined as

$$\Omega^t = \begin{pmatrix} \vdots \\ \Phi^t_{\ell,n} \\ \vdots \\ \phi^t_s \\ \vdots \\ \Phi^t_{\mathrm{g},n} \\ \vdots \end{pmatrix}, \quad \tilde{f}^t_0 = \begin{pmatrix} \vdots \\ \tilde{f}^t_{0\ell,n} \\ \vdots \\ \tilde{f}^t_{0s} \\ \vdots \\ \tilde{f}^t_{0\mathrm{g},n} \\ \vdots \end{pmatrix}, \quad n = \{1, ..., \mathrm{n}_\beta\}, \tag{2.32}$$

The normalized residuals vector

A solution obtained by the iterative procedure would be considered as converged after $k+1$ iterations if $\left\| \tilde{f}^{k+1}_0 \right\|_\infty \leq \mathrm{tol}$. However, the vector $\tilde{f}_0$ is made of terms of very different magnitude as it includes the numerical error of the enthalpy and mass fractions equation in its definition, with numerical values of very different magnitude integrated over different volumes and areas. To facilitate the comparison between the convergence histories, each of the elements of this vector is normalized to obtain the scaled error vector f_0, defined by dividing $\tilde{f}_0$ with the value of the variables at a previous time $V^{t-\Delta t}_n$, $\rho^{t-\Delta t}_{\beta,n}$ and $h^{t-\Delta t}_{\beta,n}$.
The iterative procedure described here can be stretched to reach normalized residuals tolerances as stringent as $\|f_0\|_\infty < \mathrm{tol} = 10^{-12}$, but a tolerance $\mathrm{tol} = 10^{-8}$ is usually enough to achieve a sufficiently accurate solution.

The numerical problem reduces then to minimize the error vector $\|f_0\|_\infty < \mathrm{tol}$ and to find the location of the interface a at every time step, a stiff nonlinear Differential-Algebraic Equation system (DAEs) in which the mesh evolution is coupled with the flow field solution. To that end, we used the iterative Stepped Shamankii method to obtain the numerical value of the variables $\Omega^{t+\Delta t}$ at a generic time $t+\Delta t$. For simplicity in the notation, in what follows we will drop the super-index $t+\Delta t$, so that $\Omega^k_\beta = \Omega^{t+\Delta t,k}_\beta$.

The Stepped Shamanskii Method (SSM) is based on the Shamankii method described by [32–34] but it is modified to introduce three different damping coefficients for the liquid phase, the interface and the gas phase λ_β, $\beta = \{\ell, \mathrm{g}, s\}$, respectively. The solution is updated according to

$$\Omega^{k+1}_\beta = \Omega^k_\beta - \lambda^k_\beta \Delta\Omega^{k+1}_\beta \tag{2.33}$$

where Ω^{k+1}_β is the $(k+1)$-th approximation to the unknowns vector, at $t+\Delta t$, for the β phase and $\Delta\Omega^{k+1}_\beta$ are the elements of the vector $\Delta\Omega^{k+1} = J^{-1}\Omega^k$ that correspond to the variables of the phase β. The Jacobian matrix J is updated

every $p = 20$ iterations to reduce the computational cost, as described in [32–34]. We defined a damping coefficient for every fluid phase, such that

$$\lambda_\beta^k = \lambda^k \Lambda_\beta^k \tag{2.34}$$

with the phase-dependent, step-damping coefficient Λ_β updated every p iterations as

$$\Lambda_\beta^k = \min(r_\beta \Lambda_\beta^{k-1}, 1), \quad 0 \leq \Lambda_a \leq \Lambda_\mathrm{g} \leq \Lambda_\ell \leq 1 \tag{2.35}$$

and $r_\beta = \{1, 2, 10\}$ is the step expansion ratio for the $\beta = \{\ell, \mathrm{g}, s\}$ phase. The step-damping coefficients are initialized as

$$\Lambda_\beta^0 = r_\beta^{-\eta_\beta}. \tag{2.36}$$

with $\eta_\beta = \{1, 2, 3\}$ the step order sequence parameter. This modification allows the liquid phase to converge first, followed by the gas phase and, finally, the interface. In this modification, the interaction between the interface and the reaction zone, separated by a distance of the order of a few radii, are smoothed and the method provides good convergence properties. Notice that the SSM briefly described above reduces to the well-known Newton Method when the damping coefficient λ^k is chosen to be constant and the Jacobian matrix J is updated every iteration. A simplified sketch of the Stepped Shamanskii method flowchart is presented in figure 2.3, together with the different algorithms involved in the global iteration.

To ensure the convergence of the iterative method, the Armijo rule [34] has been used. The damping parameter $0 < \lambda^k < 1$ is selected to guarantee that the condition

$$\left\| f_0^{k+1} \right\|_\infty \leq \left(1 - \gamma \lambda^k\right) \left\| f_0^k \right\|_\infty \tag{2.37}$$

is satisfied with $\gamma = 10^{-4}$ as suggested in [34]. Even though this strategy requires of some trial and error in the selection of λ^k, it is designed to obtain convergence towards the solution of the problem as the iteration proceeds. However, sometimes, the convergence of the iterative process using the Armijo rule can be difficult to achieve, specially when the initial profile is not sufficiently accurate. For that reason, we introduced a modification to the Armijo rule that consists in smoothing the initial profiles of the variables and fixing the damping parameter for the first p iterations to a small value $\lambda^k = 10^{-2}$ until the Jacobian is updated for the first time. The successive approximation to the solution Ω^k is corrected to avoid non-physical solutions by capping the species mass fractions $0 < Y_{\beta,i} < 1$ and the maximum gas temperature $0 < T < 10,000$K. After these p initial iterations, the normal procedure of Armijo rule (2.37) is resumed.

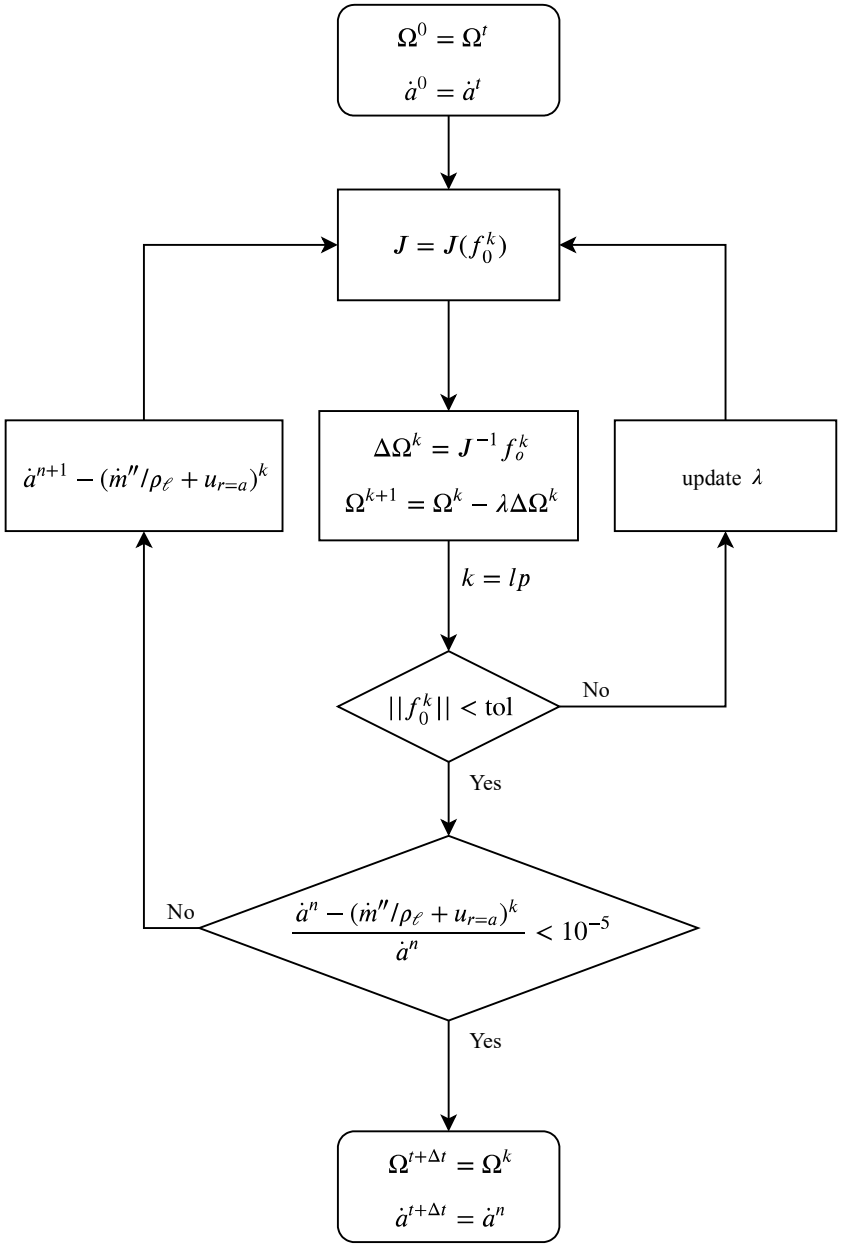


Figure 2.3: Flowchart of the iteration procedure for the Stepped Shamanskii Method and the predictor-corrector for the evaluation of $\mathrm{d}a/\mathrm{d}t$ and Ω at instant $t + \Delta t$

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The Jacobian matrix evaluation

The jacobian matrix evaluation is very expensive from a computational point of view. An appropriate balance between accuracy and efficiency is obtained using a first order approximation in the numerical evaluation of the Jacobian matrix,

$$J\left(f_{\mathrm{o}}\right)=\frac{f_{\mathrm{o}}(\Omega+\delta\Omega)-f_{\mathrm{o}}(\Omega)}{\delta\Omega}, \tag{2.38}$$

where

$$\delta\Omega=a\max(b^{-l},\varepsilon)\Omega+\epsilon \tag{2.39}$$

being $a=10^{-3}$, $b=5$, $\varepsilon=10^{-6}$ and $\epsilon=10^{-15}$ typical values used during the computations shown below, and the super-index l is a value that starts at $l=0$ and increases continuously by 1 until the Jacobian is updated. The dynamic definition of the Jacobian matrix introduced in (2.39) increased the accuracy of the method providing high computational efficiency without the increase in the number of iterations that a second order evaluation of the Jacobian matrix would introduce. The evaluation of the inverse Jacobian matrix is achieved using a LU decomposition that takes advantage of the $6(N_{\mathrm{g}}+2)$-width banded structure of the Jacobian matrix. The Shamanskii method is initialized with the solution in the previous time iteration as $\Omega^0=\Omega^t$.

2.3.5 The time stepping strategy

Linear stability analysis of ODE solvers reveals that implicit schemes are stable for arbitrary $\Delta t>0$. Given an instant t, the solution for $t+\Delta t$ could, in principle, be obtained with the only restriction of accuracy. However the time step Δt is limited by non-linearities, especially when large variations occur in a very short time interval, as occurs for instance at the sudden droplet autoignition described in chapter 5. This phenomenon requires small time steps, typically $\Delta t<10^{-5}$s, to be appropriately described. Even for such small time steps, the stiffness of the problem can make impossible to attain a converged solution using a constant time step. To circumvent this problem, one of the options is to adapt the time step to the instantaneous stiffness of the problem. The maximum time step used during the calculations is limited by the accuracy and stability of the numerical algorithm employed during the integration of the equations. On one hand, it should be sufficiently small to ensure accuracy but, on the other hand, should be as large as possible to reduce the computational cost.

Ideally, the estimation of the numerical method error should be done in real-time during the course of the simulation to keep the errors below a given threshold adapting Δt. A technique typically used in ODE problems consists in solving the problem simultaneously with two methods of different accuracy (discretization

error). The higher order method provides a solution that works as a reference solution to build a dynamic error estimation that will be used to adapt the time step. This method is used, for instance, in the Runge-Kutta-Fehlberg method [35]. Even though this is attractive, implicit numerical schemes, as the one used here to solve the evaporation and combustion of a fuel droplet, would require doubling the number of operations per time step, with the corresponding loss of computational efficiency.

For the reasons enumerated above, we measured indirectly the stiffness with the number of iterations necessary for the system to converge. A strategy to control the time step is to increase Δt when in the previous time steps convergence was easily achieved, and to reduce Δt whenever the SSM cannot attain convergence. This procedure ensures that when changes in the solution occur over long characteristic times, the number of time steps remains reasonable. On the contrary, when convergence difficulties are found, the time step is reduced.

The time step reduction strategy

After solution at time t is converged from the solution at $t - \Delta t$, the solution at a new time $t + c\Delta t$, with $c \geq 1$ is sought. If convergence criteria are not met, the algorithm returns to the converged solution at $t - \Delta t$ and finds the solution for $t - \Delta t/2$. Notice that, in this case, the iteration should converge as it successfully had done so when starting at $t - \Delta t$ with a time step Δt. In this way, the solution is approaching t with a smaller time step while a backup solution is available at $t - \Delta t/2$, closer than before.

Some comercial codes as COSILAB [36], use more aggressive time step reduction factors, but in this work we found that such drastic reductions in time step, while appropriate in finding final steady states, reduce the quality of the unsteady solution.

The time step growth strategy

When the unsteady evolution of the solution is slow and the stiffness is not large, the use a very small time step is inefficient. The algorithm will increase the time step after the q previous time steps have successfully converged without variation in the time step. An absolute maximum is also taken into account, so after q successful time steps the time step is updated as $\Delta t_{\text{new}} = \min(1.1\Delta t, \Delta t_{\text{max}})$. Typically, $q = 10$ was used.

The weakness of the strategy described above lies in the lack of a measure for the error made in each time step, which has to be checked afterwards.

2.3.6 Performance of the code

In this section we will test the numerical accuracy and consistency of the discretization.The mesh definition will be based only on the grid spacing h. In spite of the lack of an exact analytical solution, a good approximation to exact solutions can be obtained by progressively reducing h and using Richardson extrapolation; in this way, both the discretization error and the order of the approximation can be estimated as Order $= \log_2((\Phi(h) - \Phi(h/2))/(\Phi(h/2) - \Phi(h/4)))$.

Two benchmark problems will be used, one to characterize the spatial discretization and the other to characterize the time discretization:

Test 1

This first benchmark consists in the vaporization of a pure ethanol droplet of constant diameter and temperature $d_0 = 1$mm and $T_{d_0} = 300$K, respectively, in a hot nitrogen atmosphere at temperature of $T_\infty = 500$K and pressure $p_\infty = 1$bar. Steady solution of the equations for the gas phase is sought, while the liquid phase remains at the same specified initial conditions. Therefore, only the gas phase and the interface are solved until a steady solution is reached. The mesh is progressively refined in both phases close to the droplet surface. The mesh is specified by the following parameters: $r_\mathrm{I} = 3a_0$, $r_\mathrm{end} = 100a_0$, $n_\mathrm{g} = 2n_\mathrm{g,I} = 4n_\ell$ and $n_\ell = a_0/\Delta r$. The variables to be analyzed are the spatial step size, Δr, and the surface temperature, T_s; results obtained are shown in table 2.1. The order of accuracy tends to 1 as the spatial step size $\Delta r \to 0$, as should be expected from the one-sided first order discretization of heat and mass transfer at both sides of the interface (2.17), and from the first order discretization of the convective terms.

$\Delta r/a_0 \times 10$	T_s	error	Order
2	302.2992	1.0734	0.8539
1	301.2258	0.5939	0.9320
0.5	300.6319	0.3113	0.9677
0.25	300.3206	0.1592	0.9862
0.125	300.1615	0.0803	-
0.0625	300.0812	-	-

Table 2.1: Error estimation and order of accuracy for the numerical method at the interface steady solution.

Test 2

The second benchmark consists in the analysis of the temporal evolution of a pure ethanol droplet vaporization in a hot dry nitrogen atmosphere at temperature of $T_\infty = 500$K and pressure $p_\infty = 1$bar with an initial droplet diameter of $d_0 = 1$mm and temperature of $T_{d_0} = 300$K. The mesh parameters employed are defined as in

the previous cases, being: $r_{\mathrm{I}} = 3a_0$, $r_{\mathrm{end}} = 10a_0$, $n_\ell = 10$, $n_{\mathrm{g}} = 40$, and $n_{\mathrm{g,I}} = 40$. The time evolution of the problem will be controlled by the maximum time step allowed by the code, Δt_{max}, which will be the initial time step and will not be modified til the end of the problem. The simulation stops at $t_{\mathrm{end}} = 20$ms and we will analyze the fuel mass fraction at $r = 5a_0$ at the end of the simulation. Table 2.2 summarizes the results from this last test and reveal that the temporal discretization introduces errors of order $\mathcal{O}(\Delta t)$, as should be expected from a backward Euler discretization .

$\Delta t_{\mathrm{max}} \times 10^3$	$Y_{fuel} \times 10^3$	error$\times 10^4$	Order
1	1.6977	0.2230	0.9554
0.5	1.6754	0.1150	0.9386
0.25	1.6639	0.0600	0.9527
0.125	1.6579	0.0310	-
0.0625	1.6548	-	-

Table 2.2: Error estimation and order of accuracy for the numerical method for transient solution at the gas phase.

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